

COMENIUS UNIVERSITY IN BRATISLAVA
FACULTY OF MATHEMATICS, PHYSICS AND INFORMATICS

HARDWARE DESIGN AND DOCUMENTATION OF
THE UACS RFID AUDIO CAN NODE
BACHELOR THESIS

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COMENIUS UNIVERSITY IN BRATISLAVA
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HARDWARE DESIGN AND DOCUMENTATION OF THE UACS RFID AUDIO CAN NODE

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Declaration: I hereby declare that I have independently written this entire bachelor thesis entitled “Hardware Design and Documentation of the UACS RFID Audio CAN Node”, including all its figures and appendices, using the literature listed in the attached bibliography.

In preparation of this thesis I have used the Claude AI assistant for the following purposes: as a technical advisor on topics I was not previously familiar with, for spelling and formatting corrections, for suggesting sources I could use as a basis for claims, and for converting the thesis from Markdown to L^AT_EX format. I have used this tool in accordance with relevant legal regulations, academic rights and freedoms, ethical and moral principles while simultaneously maintaining academic integrity. I am aware that I am fully responsible for the accuracy of the final text.

Acknowledgments:

Abstrakt

Klíčové slová: UACS, RFID, CAN FD, STM32H523, doska plošných spojov, hardvérový návrh

Abstract

Keywords: UACS, RFID, CAN FD, STM32H523, printed circuit board, hardware design

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Introduction

Project: UACS (RFID Audio CAN Node)

Repository: <https://github.com/ostertag/UACS>

Microcontroller: STM32H523CET6 (Arm Cortex-M33, LQFP-48)

EDA tools: KiCad (schematic + PCB), STM32CubeMX (pin/peripheral assignment)

Debug hardware: STLINK-V3PWR

Component sourcing: LCSC / JLCPCB

Overview

UACS is a custom-designed printed circuit board that integrates contactless RFID reading, audio feedback through a loudspeaker, CAN FD communication, and SD-card audio storage, built around an STM32H523CET6 microcontroller. The board communicates over a CAN FD link, carried over standard RJ45/Cat5 cabling, to a host computer (a Raspberry Pi).

The STM32H523xx is a high-performance microcontroller built on the Arm Cortex-M33 32-bit core, operating at up to 250 MHz, with a single-precision FPU and TrustZone security extension [22].

System Architecture

The system is a single-board reader node whose main functional blocks are:

- **Microcontroller** — STM32H523CET6 (Cortex-M33, 250 MHz, LQFP-48), central coordinator of all peripherals.
- **RFID reader** — RC522 module (NXP MFRC522), reads ISO/IEC 14443 A / MIFARE cards and tags over SPI.
- **Audio amplifier** — MAX98357A, digital Class D amplifier with I²S input; drives the loudspeaker directly.

- **SD card** — micro-SD slot for audio file storage, connected over SPI.
- **CAN FD transceiver** — TCAN1057A, links the board to the host over Cat5/RJ45 cabling.
- **Power supply** — dual 5 V input (USB-C and RJ45), stepped down to 3.3 V by an MCP16301H buck converter.
- **USB-C interface** — USB 2.0 full-speed device, used for firmware flashing (DFU) and debugging.
- **Indicator LEDs** — two dual-colour (red/green) SMD LEDs for user feedback.

Scope and Goal of This Work

This work continues the UACS project previously developed in the Deadlock project series [6]. The principal goal is a thoroughly documented hardware design that can be reliably reproduced. In particular:

- **Reproducibility** — every component value is either calculated from first principles or traced to a specific datasheet reference, so the design can be rebuilt without reliance on undocumented institutional knowledge.
- **Availability** — components are sourced from LCSC/JLCPCB and selected for availability, good documentation, and competitive cost.
- **Robustness** — the dual power input, remote reset, and remote bootloader-entry capability allow firmware updates and recovery without physical access to a deployed board.

Chapter 1

Design and Assembly of the UACS PCB

The UACS reader is a printed circuit board on which an STM32H523CET6 microcontroller, an RFID front-end module, an I²S audio amplifier, a micro-SD slot, a CAN FD transceiver, and associated connectors are assembled. User feedback is provided by two dual-colour LEDs. The board can be powered either from a USB-C connector or from the RJ45 CAN cable.

1.1 Microcontroller — STM32H523CET6

The STM32H523CET6 is the central controller of the board. It is supplied in an LQFP-48 package and runs the Cortex-M33 core at 250 MHz [22]. All peripheral communication — SPI, I²S, FDCAN, and USB — is handled by this device.

1.1.1 Debug Interface (SWD)

The board exposes a 10-pin 1.27 mm-pitch Cortex debug connector (SWD1), driven by an STLINK-V3PWR probe.

The STM32H523 supports both JTAG and Serial Wire Debug (SWD), as well as the Embedded Trace Macrocell (ETM) [22]. SWD was selected over JTAG because it reduces the debug interface to two active signals — SWDIO and SWCLK — while providing equivalent core-debug and memory-access capability to the higher-pin-count JTAG interface [4, 20]. On a pin-constrained LQFP-48 device, freeing the JTAG pins (TDI, TDO, nTRST) for other functions is a meaningful saving, and SWD is the conventional default for Cortex-M bring-up [20].

The Serial Wire Output trace pin (TRACESWO) is brought out on PB3, enabling single-wire instrumentation trace (printf-style debug output) without consuming

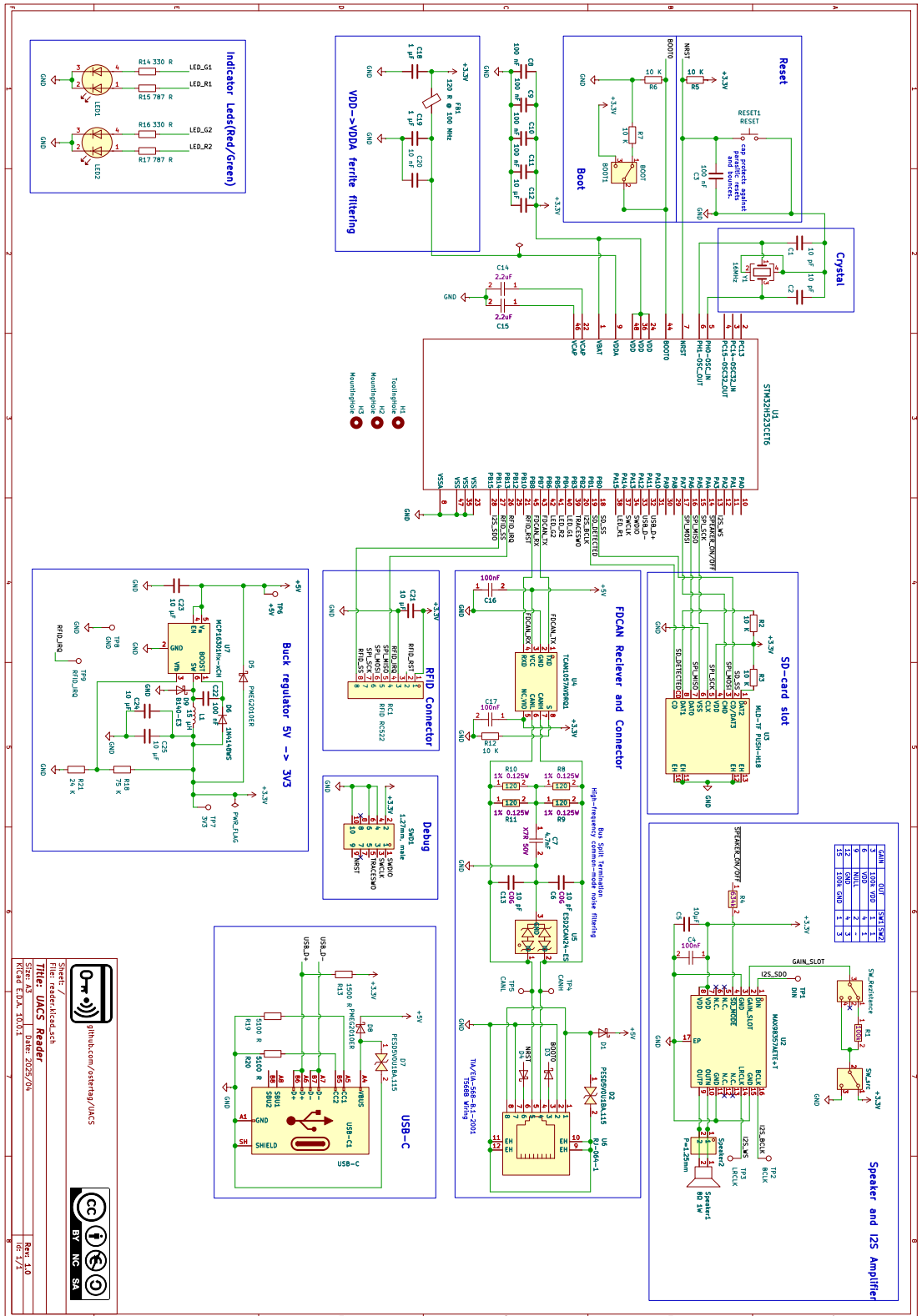


Figure 1.1: UACS board full schematic.

a UART peripheral. The connector follows the standard 10-pin 1.27 mm Cortex-debug pinout, which balances size against reliability [20].

The STLINK-V3PWR probe as we already had it available [21].

1.1.2 Reset Circuit

The NRST line is shared between three reset sources: the on-board push button (RESET1), the remote reset line carried over the RJ45 connector (Section 1.4.2), and the STLINK debug probe. If software reset over SWD fails, the probe can still assert NRST to force a hardware reset [20].

Resistor R5. A 10 k Ω pull-up resistor holds the NRST line high during normal operation. Without it the line would float and could trigger spurious resets from noise.

Capacitor C3. A 100 nF capacitor from NRST to GND provides debounce for the mechanical reset button and protection against parasitic resets. ST explicitly recommends an external capacitor on NRST for exactly this purpose [22].

1.1.3 BOOT0 Circuit

BOOT0 selects whether the MCU starts from user flash (normal operation) or from the embedded system bootloader. The STM32H523 embedded bootloader, located in system memory and programmed by ST during production, can reprogram the flash over USART, I²C, I³C, SPI, FDCAN, or USB-FS (via DFU) [22]. On UACS, USB-FS DFU is the relevant path (Section 1.6).

BOOT0 is additionally exposed on the RJ45 connector (Section 1.4.2), so the bootloader can be entered remotely without physical access to the board.

Resistor R6. A 10 k Ω pull-down resistor holds BOOT0 low during normal operation. With BOOT0 low, the device boots from user flash at address 0x0800 0000 [22].

Resistor R7. A 10 k Ω series resistor is placed between the R6 junction and the BOOT1 switch to limit current and provide isolation between the pull-down and the switch.

Switch BOOT1. An SPDT switch that selects GND (normal boot) or 3.3 V (bootloader entry). Connecting BOOT0 to 3.3 V causes the MCU to enter the embedded system bootloader on the next reset.

1.1.4 Crystal Oscillator

A 16 MHz crystal (Y1, SMD3225-4P, 10 pF load) provides the high-speed external (HSE) clock. The STM32H523 supports a 4–50 MHz HSE oscillator [22]. Although the device has an internal 64 MHz HSI oscillator, an external crystal is used because the USB and CAN peripherals require a more accurate and stable time base than the internal RC oscillator can guarantee. The USB full-speed interface has particularly tight frequency-tolerance requirements.

No 32.768 kHz crystal is fitted; the LSE pins (PC14/PC15) are left unconnected and configured in Analog mode.

Capacitors C1 and C2. Two 10 pF C0G load capacitors are placed to GND, one on each side of the crystal, matched to its specified 10 pF load capacitance. C0G (Class 1) dielectric is used because its capacitance is essentially independent of temperature and applied voltage — typically within ± 30 ppm/°C — which is important for a stable oscillator load [11, 1].

1.1.5 VCAP — Internal Regulator Capacitors

The STM32H5 family integrates an LDO voltage regulator (nominally $3.3\text{ V} \rightarrow 1.2\text{ V}$ core) that requires external stabilisation capacitors on its VCAP pin(s) [22]. The datasheet specifies this requirement explicitly in Section 5.3.2, “VCAP external capacitor” [22].

Capacitors C14 and C15. Two $2.2\text{ }\mu\text{F}$ capacitors, one per VCAP pin, are placed to GND. These are mandated by the datasheet for the internal regulator’s stability and are not interchangeable with generic decoupling — the value and placement are part of the regulator’s stability requirement.

1.1.6 Power Pins and Decoupling

The STM32H523 uses several supply domains: the main VDD digital supply, an analog supply (VDDA), a USB supply domain, and the VBAT backup domain [22].

- **VDD** — supplied from $+3.3\text{ V}$; decoupled with C8–C11 (100 nF each) for high-frequency bypass, plus C12 (10 μF) for bulk capacitance.
- **VDDA** — analog supply, filtered from the 3.3 V rail through the VDDA filter (Section 1.3.3).
- **VBAT** — tied to the 3.3 V rail, since no backup battery is used.
- **VSS / VSSA** — ground.

1.1.7 RFID Reader — RC522 Module (MFRC522)

The RFID front end is an RC522 module based on the NXP MFRC522 IC, mounted on an 8-pin 2.54 mm socket (RC1) and connected to the MCU over SPI.

The RC522 module was selected because it was used in the predecessor project, allowing existing experience and firmware to be carried over. The MFRC522 is a 13.56 MHz reader/writer IC supporting the ISO/IEC 14443 A / MIFARE protocol family — MIFARE Mini, 1K, 4K, Ultralight, DESFire EV1, MIFARE Plus, and NTAG — with a host SPI interface running up to 10 Mbit/s [18].

Interface signals:

- **SPI** — shares SPI1 with the SD card (Section 1.2); chip select is software-controlled on PB14 (RFID_SS), active low.
- **RST (PB10)** — the MCU drives the module's NRSTPD pin as a push-pull output. The pin defaults high at boot so the reader starts in normal operation; driving it low triggers a hard reset/power-down of the module [18].
- **IRQ (PB13)** — the module's interrupt output drives EXTI13, configured for falling-edge interrupts, enabling interrupt-driven card detection rather than polling.

On the MFRC522, the IRQ pin is open-drain and active-low by default [18], which matches the MCU's falling-edge EXTI configuration; only an MCU-side internal pull-up is required on PB13, with no external resistor. The MFRC522's CommIEnReg register must be configured in firmware to enable interrupt propagation to the IRQ pin [18].

Capacitor C21. A 10 μ F local bypass capacitor is placed on the module VCC pin. VCC connects directly to the 3.3V rail. (PB12 is left unconnected; an earlier plan to use it for switched module power was dropped, since a single GPIO cannot source the MFRC522's worst-case current.)

1.2 SD Card — MLD-TF PUSH-H18

The micro-SD slot (U3) stores the audio files played back through the amplifier.

The preferred interface would have been the dedicated SDMMC peripheral, which provides higher throughput than SPI. However, the SDMMC peripheral on the STM32H523 is only available on larger packages; on the LQFP-48 the required pins are not bonded out, so SDMMC is physically unavailable and SPI is the only option [22]. This is a package constraint, not a design preference.

The SD card shares SPI1 with the RC522 reader, with its own software chip select on PB0 (SD_SS). In SPI mode, the card's CD/DAT3 pin (slot pin 2) functions as the

active-low chip select; this is confirmed by the slot’s pin table, which labels pin 2 as CD/DAT3 with I/O type PP [19].

The MLD-TF PUSH-H18 uses a push-push ejection mechanism (card ejects on the second press); the card-detect switch is a separate mechanical contact from the signal pins [19].

Card Detect — PB1. The slot’s dedicated card-detect mechanical contact drives PB1 (SD_DETECTED) as a GPIO input, allowing the firmware to detect card insertion and removal.

Resistors R2 and R3. Two 10 k Ω pull-up resistors are placed to 3.3 V on the DAT1 line (R2) and the DAT2 line (R3). In SPI mode both of these data lines are unused; the pull-ups keep them at a defined high level rather than floating.

1.3 Power Supply

1.3.1 Architecture

The board accepts power from two independent 5 V sources, diode-OR-ed onto a common +5 V rail and then stepped down to 3.3 V by a buck converter.

Table 1.1: Power input OR-ing and ESD protection.

Source	OR-ing diode	ESD protection
USB-C VBUS	D8 (PMEG2010ER)	D7 (PESD5V0U1BA,115)
RJ45 pair (brown)	D1 (PMEG2010ER)	D2 (PESD5V0U1BA,115)

The two inputs serve different operational scenarios. The RJ45 input is the primary supply during normal deployment, where the board is powered from the CAN cabling infrastructure. The USB-C input is used during development and programming, when the board is connected to a PC without the full CAN harness. Diode OR-ing lets both sources be connected simultaneously without back-feeding. Each TVS diode is placed as close as possible to its respective input connector for effective clamping [17].

1.3.2 Buck Converter — MCP16301H

The MCP16301H (U7) is a high-voltage-input, integrated-switch, fixed-frequency step-down converter in a SOT-23-6 package [13]. It is AEC-Q100 Grade 1 automotive-qualified and can supply up to 1 A at 3.3 V output [13]. The part was carried over from the previous version of the design, where it was already proven.

The UACS buck stage follows the MCP16301H datasheet’s own 3.3 V reference design very closely, which is a deliberate choice for a known-good result.

Inductor L1. A 15 μH inductor is used. The datasheet explicitly recommends 15 μH for a 3.3 V output, derived from the inductor-selection constant K :

$$K = \frac{V_{\text{OUT}}}{L} \quad (1.1)$$

where K should equal $0.22 \text{ V } \mu\text{H}^{-1}$, giving:

$$L = \frac{V_{\text{OUT}}}{K} = \frac{3.3}{0.22} \approx 15 \mu\text{H} \quad (1.2)$$

Table 1.2: MCP16301H inductor selection for common output voltages [13].

V_{OUT} (V)	K (V/ μH)	Standard L
2.0	0.20	10 μH
3.3	0.22	15 μH
5.0	0.23	22 μH
12.0	0.21	56 μH
15.0	0.22	68 μH

Diode D9 (B140-E3, SMA). Schottky freewheeling diode; the datasheet reference design uses the B140 [13].

Diode D6 (1N4148WS). Boost (bootstrap) diode. The datasheet specifically recommends a 1N4148 for this role for its fast recovery speed, voltage blocking capability, wide availability, and low cost [13].

Capacitor C22. A 100 nF boost capacitor. The datasheet recommends a 0.1 μF X5R/X7R capacitor for all applications [13].

Capacitors C23, C24, C25. C23 (10 μF) is the input capacitor; C24 and C25 (10 μF each) are output capacitors placed in parallel. Their values are consistent with the datasheet’s recommended input/output capacitance [13].

Diode D5 (PMEG2010ER). Back-powering protection diode between the 3.3 V output and the 5 V input, reverse-biased during normal operation. It prevents the 3.3 V rail from back-feeding the 5 V input when only the 3.3 V side is powered — for example, when supplied from the STLINK probe during development.

Feedback Divider (R18 and R21) — Output Voltage Setting. The MCP16301H regulates to an internal feedback reference of **0.800 V (typical)**. The output voltage is set by an external resistor divider according to the datasheet formula [13]:

$$R_{\text{TOP}} = R_{\text{BOT}} \times \left(\frac{V_{\text{OUT}}}{V_{\text{FB}}} - 1 \right) \quad (1.3)$$

equivalently:

$$V_{\text{OUT}} = V_{\text{FB}} \times \left(1 + \frac{R_{\text{TOP}}}{R_{\text{BOT}}} \right) \quad (1.4)$$

With the values used on UACS ($R_{18} = 75 \text{ k}\Omega = R_{\text{TOP}}$, $R_{21} = 24 \text{ k}\Omega = R_{\text{BOT}}$):

$$V_{\text{OUT}} = 0.800 \times \left(1 + \frac{75}{24} \right) = 0.800 \times 4.125 \approx 3.30 \text{ V} \quad \checkmark \quad (1.5)$$

The datasheet's own example 3.3 V divider uses $R_{\text{TOP}} = 31.6 \text{ k}\Omega$ and $R_{\text{BOT}} = 10 \text{ k}\Omega$ [13]. UACS uses higher-value resistors ($75 \text{ k}\Omega / 24 \text{ k}\Omega$); both ratios produce 3.3 V. The higher values were carried over from the predecessor design, where they were chosen to reduce quiescent current through the divider.

1.3.3 VDDA Filter

The MCU analog supply (VDDA) is filtered from the digital 3.3 V rail to keep switching noise out of the analog domain (ADC reference, internal oscillator). A ferrite bead in series with the supply, combined with decoupling capacitors on each side, forms a low-pass filter that absorbs high-frequency noise and dissipates it as heat; this is the standard technique for isolating a sensitive analog supply from a noisier digital rail [2].

Ferrite Bead FB1. A 120Ω @ 100 MHz, 500 mA, 0603 ferrite bead acts as the series filter element, presenting high impedance to high-frequency noise while passing DC with negligible voltage drop.

Capacitor C18. A $1 \mu\text{F}$ capacitor placed before the ferrite bead, from the digital 3.3 V rail to GND. Together with the bead, it forms the input side of the low-pass filter.

Capacitors C19 and C20. A $1 \mu\text{F}$ and a 10 nF capacitor placed after the ferrite bead, from the VDDA node to GND. The combination of C19 and C20 provides filtering across a wide frequency range, covering both mid-frequency and high-frequency noise at the filtered VDDA node.

1.4 CAN Bus — TCAN1057A

The CAN interface links the board to the host Raspberry Pi. The transceiver is the TCAN1057AVDRQ1 (U4), an automotive-grade CAN FD device in a SOIC-8 package, driven by the MCU FDCAN1 peripheral (PB7 TX, PB8 RX) [24].

The TCAN1057A was selected because it meets the required parameters, including CAN FD support. CAN FD (Flexible Data-rate) extends classical CAN by allowing the data phase to be transmitted at a higher bit rate than the arbitration phase, and by increasing the maximum payload from 8 to 64 bytes per frame [12, 15]. In practice CAN FD data-phase rates of up to 5 Mbit/s are common [15], giving the link more bandwidth and headroom than classical CAN, while remaining compatible with the familiar CAN arbitration and error-handling mechanisms [12].

1.4.1 Transceiver Configuration

Capacitors C16 and C17. Two 100 nF bypass capacitors are placed on the transceiver VCC and GND pins for high-frequency supply decoupling.

Resistor R12 — Mode Select Pin. A 10 k Ω resistor ties the S (mode-select) pin permanently to GND, placing the transceiver in Normal Mode. The datasheet (Table 8-4) confirms: S = Low $\rightarrow$ Normal Mode (driver and receiver both enabled); S = High $\rightarrow$ Silent Mode (driver disabled, receive-only). The datasheet explicitly states that when normal mode is the only intended operating mode, the S pin may be tied directly to GND via a pull-down resistor [24].

1.4.2 RJ45 Cabling and Pair Assignment

The bus is carried over standard RJ45/Cat5 cabling. The eight conductors are assigned as four pairs:

Table 1.3: RJ45 connector pair assignments.

Connector pins	Pair function
1 + 2	GND / +5 V (power)
3 + 6	BOOT0 / NRST (remote MCU control)
4 + 5	CANH / CANL (differential bus)
7 + 8	GND / +5 V (power)

In a Cat5 cable, pins 4 and 5 form the centre, tightest-twisted pair. The differential CAN signal is assigned to this pair because tight twisting maximises noise rejection:

both conductors pick up external interference almost equally as common-mode, which the differential receiver then rejects.

Carrying BOOT0 and NRST over the cable allows remote firmware flashing and reset without physical access to the deployed board.

Diodes D3 and D4 (PMEG2010ER). Protection diodes placed on the BOOT0 (D3) and NRST (D4) lines respectively, guarding against overvoltage transients on these control signals.

1.4.3 Bus Termination

Termination is the most involved part of the CAN design and is documented in full below.

Why termination is required. A CAN bus is a transmission line whose characteristic impedance ($\approx 120\,\Omega$ for twisted-pair CAN cable) must be matched at both ends. Without $120\,\Omega$ termination at each end, signal edges reflect from the unterminated ends and corrupt communication. Terminating both ends absorbs the signal energy and prevents these reflections [8].

Why split termination instead of a single $120\,\Omega$ resistor. In a split-termination scheme the single $120\,\Omega$ resistor is replaced by two $60\,\Omega$ resistors in series, with a capacitor from their midpoint to ground. The differential impedance remains $120\,\Omega$ ($60\,\Omega + 60\,\Omega$), so CAN signalling is unaffected, while the midpoint capacitor provides a low-impedance path to ground for high-frequency common-mode noise — improving electromagnetic compatibility (EMC) [5, 23]. This is particularly valuable on a board that also carries a switching regulator and a multi-metre cable run.

Why two $120\,\Omega$ resistors in parallel per side. Each $60\,\Omega$ half is implemented using two $120\,\Omega$ resistors in parallel, rather than a single $60\,\Omega$ part. Single $60\,\Omega$ resistors in the 0805 footprint are less commonly stocked and more expensive; $120\,\Omega$ is a very common value, and two in parallel give exactly $60\,\Omega$. A secondary benefit is that power dissipation is split across two resistors, halving the dissipation each must handle.

Resistors R8 || R9 (CANH side). Two $120\,\Omega$ resistors in parallel, giving $60\,\Omega$, placed in series on CANH. Each resistor is 0805, 1%, 0.125 W.

Resistors R10 || R11 (CANL side). Two $120\,\Omega$ resistors in parallel, giving $60\,\Omega$, placed in series on CANL. Same specification as R8/R9.

Capacitor C7. A $4.7\,\text{nF}$ X7R, 50 V capacitor from the split-termination midpoint to GND. The chosen value sits within the commonly recommended 4.7–10 nF range for common-mode filtering [5].

1.4.4 High-Frequency Bypass and ESD Protection

Capacitors C6 and C13. Two 10 pF C0G, 50 V capacitors are placed from CANH (C6) and CANL (C13) respectively to GND, providing additional high-frequency bypass on the bus lines. C0G (Class 1) dielectric is specified deliberately: its capacitance is stable with voltage and temperature, whereas a Class 2 dielectric such as X7R varies with both and would risk rounding the fast CAN FD signal edges [11, 1].

U5 — ESD Protection (ESD2CAN24-ES, SOT-23-3). A CAN-specific TVS/ESD protection device placed on the bus side of the filtering network. The device is rated for ± 30 kV IEC 61000-4-2 contact and air discharge, with a 24 V reverse standoff voltage and a typical junction capacitance of 18 pF, making it well-suited for a CAN FD bus [7].

1.5 Audio Amplifier — MAX98357A

The MAX98357A (U2) is a digital-input Class D audio amplifier in a TQFN-16 package, supplied from 3.3 V. It receives audio over I²S and drives the loudspeaker directly.

The MAX98357A was selected for several reasons: it integrates an I²S receiver and a Class D amplifier in a single device, eliminating the separate DAC/translator, amplifier, and output-filter stages that a discrete audio path would require; it drives the speaker directly with no external output filter; it provides hardware-selectable gain; and it comes from a reputable manufacturer with thorough documentation [3].

The amplifier connects to the MCU I²S peripheral over three lines:

- **DIN — PB15 (I2S_SDO)** — serial audio data
- **BCLK — PB2 (I2S_BCLK)** — bit clock
- **LRCLK — PA3 (I2S_WS)** — word select / left-right clock

The exposed pad (EP) is tied to GND. The speaker connects through a Molex 1.25 mm right-angle connector (Speaker2).

1.5.1 SD_MODE / Shutdown Control

The SD_MODE pin controls both shutdown and the output channel mode. It is driven from PA4 (SPEAKER_ON/OFF) as a push-pull output. PA4 defaults low at boot, so the amplifier starts in shutdown.

Resistor R4 — SD_MODE Series Resistor. The output channel mode is set by the value of the resistor placed between the GPIO and the SD_MODE pin. Per the MAX98357A datasheet formula [3]:

$$R = 222.2 \times V_{DDIO} - 100 \quad [\text{result in k}\Omega] \quad (1.6)$$

Substituting the 3.3 V supply:

$$R = 222.2 \times 3.3 - 100 \approx 634 \text{ k}\Omega \quad (1.7)$$

A 634 k Ω $\pm 1\%$ resistor (R4) is used. With this configuration:

- **GPIO HIGH** (3.3 V) $\rightarrow$ SD_MODE pulled high through R4, placing the amplifier in $(L/2 + R/2)$ mono-mix mode (active).
- **GPIO LOW** (0 V) $\rightarrow$ amplifier in shutdown / deep sleep.

1.5.2 Gain Configuration (GAIN_SLOT)

The amplifier gain is set by how the GAIN_SLOT pin is connected. The datasheet [3] defines five discrete settings (Table 8):

Table 1.4: MAX98357A gain settings [3].

GAIN_SLOT connection	Gain (dB)
100 k Ω to GND	15
Direct to GND	12
Unconnected	9
Direct to VDD	6
100 k Ω to VDD	3

To make all five settings selectable in hardware, the board uses a two-switch network:

- **SW1 (SP3T)** — selects the path as open (NC), direct (0 Ω), or through a 100 k Ω resistor.
- **SW2 (SPDT)** — selects whether that path connects to GND or VDD.

The default state (GAIN_SLOT unconnected) gives **9 dB**.

Resistor R1. A 100 k Ω gain-setting resistor used for the 3 dB and 15 dB settings. Its value is taken directly from the datasheet gain table [3].

1.5.3 Speaker and Power Budget

An $8\,\Omega$, 1 W speaker is used. The output power into the load is estimated from the supply voltage and load resistance:

$$P = \frac{V_{DD}^2}{2 \times R_L} = \frac{3.3^2}{2 \times 8} \approx 0.68\,\text{W} \quad (1.8)$$

An $8\,\Omega$ load was chosen deliberately over $4\,\Omega$. At $4\,\Omega$, the output power and corresponding current draw roughly double, which would push the peak current close to the 1 A limit of the single buck regulator supplying the whole board. The $8\,\Omega$ choice keeps the peak current comfortably within budget.

1.5.4 Decoupling

Capacitors C4 and C5. C4 (100 nF) and C5 (10 μ F) provide supply decoupling on the amplifier VDD pins, covering both high-frequency and bulk requirements.

1.6 USB-C Interface

The USB-C interface (USB-C1) is a 14-pin horizontal SMD Type-C receptacle used to connect the board to a PC.

The interface is USB 2.0 only — carried over from the previous version; there is no requirement for higher transfer rates in this application. The board operates purely as a USB device (USB_DRD_FS), used for firmware flashing (USB-FS DFU) and debugging; it does not act as a host. The STM32H523 provides one USB 2.0 full-speed device interface [22].

The USB 2.0 differential data pair uses pins PA11 (D+) and PA12 (D−).

The USB-C circuit follows the standard UFP (upstream-facing port / device) topology as specified in the USB Type-C specification [9, 14].

Resistors R19 and R20. Two $5.1\,\text{k}\Omega$ Rd pull-down resistors are placed on CC1 and CC2 to GND. The USB Type-C specification requires a device (UFP) to present a $5.1\,\text{k}\Omega \pm 10\%$ Rd pull-down on each CC pin; this is the only acceptable value for advertising a standard Type-C current contract [9, 14]. The two pins are pulled down independently — not tied together [14].

Resistor R13. A $1.5\,\text{k}\Omega$ resistor is placed between the 3.3 V rail and VBUS for VBUS sensing.

Diode D7 (PESD5V0U1BA,115). A bidirectional TVS diode on VBUS for ESD protection. The device provides a 5 V reverse standoff voltage, 2.9 pF typical capacitance, and up to 10 kV IEC 61000-4-2 protection in a SOD323 package [17].

Diode D8 (PMEG2010ER). An OR-ing Schottky diode feeding VBUS into the shared +5 V power rail [16].

SBU1 and SBU2 pins are left unconnected, as they are not required for USB 2.0 device operation. The shield and GND (A1) are tied to GND.

1.7 Indicator LEDs — APHBM2012 Series

The board carries two dual-colour (red + emerald-green) SMD indicator LEDs from the Kingbright APHBM2012 series the APHBM2012LSURKZGKC. Each LED has independent red and green elements with common cathodes to GND, so each colour is driven through its own current-limiting resistor from a GPIO.

Table 1.5: Indicator LED assignments.

Ref	Nominal current	Green signal	Red signal
LED1	2 mA	LED_G1 (PB4)	LED_R1 (PA15)
LED2	2 mA	LED_G2 (PB6)	LED_R2 (PB5)

The predecessor board used two different LED variants (2 mA and 20 mA) because the required brightness was unknown during the initial design. Once confirmed that 2 mA provides sufficient visibility for the application, both LEDs were standardised to the low-current APHBM2012LSURKZGKC variant, simplifying the BOM.

Each LED element is driven from a GPIO through a series current-limiting resistor, with the cathode tied to GND.

Current-Limiting Resistors. The current-limiting resistor value is calculated from the supply voltage, the LED forward voltage, and the target forward current:

$$R = \frac{V_{DD} - V_f}{I_f} \quad \text{with } V_{DD} = 3.3 \text{ V} \quad (1.9)$$

Using the typical forward voltages from the Kingbright datasheets [10]. For LED1 with $I_f = 2 \text{ mA}$ (V_f : Red 1.75 V, Green 2.65 V):

$$R_{\text{red}} = \frac{3.3 - 1.75}{0.002} = 775 \Omega \quad \rightarrow \quad 787 \Omega \text{ chosen (R15)} \quad (1.10)$$

$$R_{\text{green}} = \frac{3.3 - 2.65}{0.002} = 325 \Omega \quad \rightarrow \quad 330 \Omega \text{ chosen (R14)} \quad (1.11)$$

Both LEDs are identical, so LED2 uses the same resistor values: R17 = 787 Ω (red) and R16 = 330 Ω (green). In both cases the nearest standard resistor value was selected.

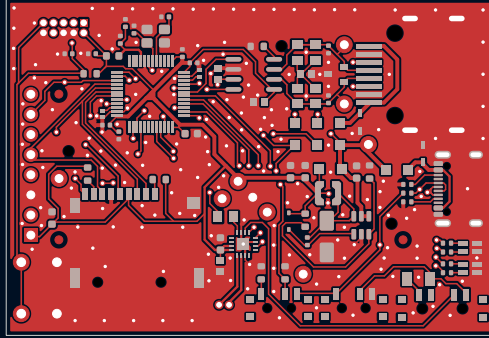


Figure 1.2: UACS PCB — front copper layer (F.Cu).

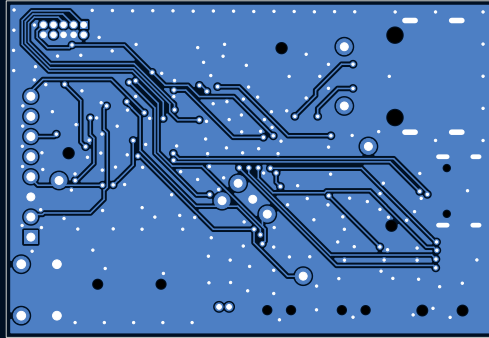


Figure 1.3: UACS PCB — back copper layer (B.Cu).

Chapter 2

Resistor Value Summary

The following table lists all resistors on the UACS board. Every value is either calculated from first principles or traced to a specific datasheet reference. This table is intended as a quick-reference companion to Chapter 1.

Table 2.1: UACS resistor value summary.

Ref(s)	Value	Basis / Derivation
R1	100 k Ω	MAX98357A GAIN_SLOT setting resistor; value taken directly from datasheet Table 8 [3].
R2, R3	10 k Ω	SD SPI-mode pull-ups on DAT1 and DAT2; lines unused in SPI mode, pull-ups prevent floating (standard practice).
R4	634 k Ω	MAX98357A SD_MODE mono-mix resistor; $R = 222.2 \times V_{DDIO} - 100 = 222.2 \times 3.3 - 100 \approx 634 \text{ k}\Omega$ [3].
R5	10 k Ω	NRST pull-up; prevents line floating; value per ST recommendation [22].
R6	10 k Ω	BOOT0 pull-down; holds BOOT0 low (user-flash boot) during normal operation [22].
R7	10 k Ω	BOOT0 series resistor; limits current and isolates pull-down from BOOT1 switch.
R8, R9	120 Ω	CAN split termination CANH side; $R8 \parallel R9 = 60 \Omega$; two 120 Ω in parallel preferred over one 60 Ω [8, 5].
R10, R11	120 Ω	CAN split termination CANL side; same rationale as R8/R9 [8, 5].
R12	10 k Ω	TCAN1057A S-pin pull-down; places transceiver in Normal Mode permanently [24].
R13	1.5 k Ω	USB VBUS sense resistor from 3.3 V rail to VBUS.

(Table 2.1 continued)

Ref(s)	Value	Basis / Derivation
R14	330 Ω	LED1 green current-limiting resistor; $R = (3.3 - 2.65) / 0.002 = 325 \Omega$, nearest E-series value 330 Ω [10].
R15	787 Ω	LED1 red current-limiting resistor; $R = (3.3 - 1.75) / 0.002 = 775 \Omega$, nearest E-series value 787 Ω [10].
R16	330 Ω	LED2 green current-limiting resistor; same derivation as R14 [10].
R17	787 Ω	LED2 red current-limiting resistor; same derivation as R15 [10].
R18	75 k Ω	MCP16301H feedback divider top resistor (R_{TOP}); together with R21 sets output to 3.3 V: $V_{\text{OUT}} = 0.8 \times (1 + 75/24) \approx 3.30 \text{ V}$; values carried over from predecessor design [13].
R19, R20	5.1 k Ω	USB-C CC1 and CC2 Rd pull-downs; mandated by USB Type-C specification for UFP (device) role [9, 14].
R21	24 k Ω	MCP16301H feedback divider bottom resistor (R_{BOT}); see R18 [13].

Chapter 3

Summary of Design Decisions

This chapter summarises the key design decisions made during the development of the UACS board and provides the technical reasoning behind each choice. The goal is to make the design intent transparent so that future maintainers can understand not only *what* was chosen, but *why*.

Table 3.1: Design decisions and their rationale.

Decision	Reasoning
STM32H523CET6 (LQFP-48)	Cortex-M33 at 250 MHz; central controller with all required peripherals (SPI, I ² S, FDCAN, USB) in a compact 48-pin package [22].
SWD debug interface (not JTAG)	Two-signal interface (SWDIO, SWCLK) provides equivalent debug capability while freeing TDI, TDO, and nTRST pins for other functions — a meaningful saving on a pin-limited LQFP-48 [4, 20].
SPI for SD card (not SD-MMC)	The SDMMC peripheral requires pins that are not bonded out on the LQFP-48 package; SPI is the only physically available option [22]. Package constraint, not design preference.
RC522 RFID module	Carried over from the predecessor design; existing firmware and proven compatibility reduce integration risk [18].
MAX98357A audio amplifier	Integrates I ² S receiver, Class D amplifier, and direct speaker drive in one device, eliminating a separate DAC, amplifier, and output filter. Hardware-selectable gain via GAIN_SLOT [3].

(Table 3.1 continued)

Decision			Reasoning
8Ω speaker (not 4Ω)			Halves the output current relative to a 4Ω load, keeping peak current within the 1 A budget of the single buck regulator that supplies the entire board.
MCP16301H buck converter			Carried over from the predecessor design; UACS follows the datasheet 3.3 V reference design closely for a known-good result [13].
Dual power input (USB-C + RJ45)			RJ45 powers the board during normal deployment from the CAN infrastructure; USB-C powers it during development and firmware flashing without requiring the full CAN harness. Diode OR-ing allows simultaneous connection.
TCAN1057A transceiver	CAN	FD	Meets parameters including CAN FD support; higher data rate (up to 5 Mbit/s data phase) and 64-byte payload compared with classical CAN [12, 15].
Split CAN termination			Maintains 120Ω differential impedance while adding a midpoint capacitor that provides a low-impedance path to ground for common-mode noise, improving EMC on a board with a switching regulator and long cable run [8, 5, 23].
Two 120Ω resistors in parallel per CAN side			60Ω resistors in 0805 are costly and less commonly stocked; 120Ω is a standard value and two in parallel give exactly 60Ω . Power dissipation is also split between two components.
C0G capacitors on CAN bus lines and crystal circuit			C0G (Class 1) capacitance is stable with voltage and temperature (± 30 ppm/ $^{\circ}\text{C}$), unlike Class 2 (X7R) which varies with both. Stability is critical for oscillator load and for preserving fast CAN FD signal edges [11, 1].
Ferrite-bead VDDA filter			Isolates the MCU analog supply from digital switching noise, protecting the ADC reference and internal oscillator. A ferrite bead + decoupling capacitor network forms a low-pass filter that dissipates high-frequency noise as heat [2].

(Table 3.1 continued)

Decision	Reasoning
USB 2.0 device only (no USB 3.x)	Carried over from the predecessor design; USB 2.0 full-speed is sufficient for DFU firmware flashing and debugging. No higher transfer rate is required in this application.
External 16 MHz crystal	USB full-speed and CAN FD require a frequency accuracy that the internal 64 MHz HSI RC oscillator cannot guarantee. An external crystal provides the necessary stability [22].
VCAP 2.2 μ F capacitors	Mandated by the STM32H5 datasheet for stability of the internal LDO voltage regulator (3.3 V $\rightarrow$ 1.2 V core); placement and value are part of the regulator's stability requirement [22].
LED resistor values from formula	$R = (V_{DD} - V_f)/I_f$, nearest standard E-series value selected, traceable to forward-voltage data in the Kingbright datasheet [10].

Conclusion

[CONCLUSION TEXT – TO BE WRITTEN]

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